

Arnav Samal

asamal@andrew.cmu.edu | (412) 812-7576 | arnavsm.github.io | github.com/arnavsm | linkedin.com/in/arnavsamal

EDUCATION

Carnegie Mellon University - School of Computer Science <i>Master of Science in Computer Vision (Robotics Institute)</i>	Pittsburgh, PA Dec 2027
National Institute of Technology, Rourkela <i>Bachelor of Technology, Computer Science & Engineering GPA: 9.13/10 Dept Rank: 7/153</i>	Rourkela, India May 2026

SKILLS

Programming Languages: Python, C++, C, SQL
Tools: Linux, Git, Conda, Weights & Biases, Docker, Jupyter, LaTeX
Libraries/Frameworks: PyTorch, OpenCV, HuggingFace, Sci-kit Learn, SciPy, NumPy, Matplotlib, Flask
Computer Vision & ML: Vision-Language Models (VLMs), Diffusion Models, Vision Transformers, Self-Supervised Learning, Explainability (CBMs, Post-hoc XAI)

RESEARCH EXPERIENCE

IISc Bangalore Visual Computing Lab, Dept. of CDS <i>Research Intern (Advisor: Dr. Anirban Chakraborty)</i>	Bengaluru, India Jan 2026 – Jun 2026
- Engineered a distributed Federated Learning pipeline to isolate how overparameterization (up to ResNet-101) mitigates client drift, benchmarking 4 SOTA algorithms (FedAvg, FedDyn, SCAFFOLD, FedProx) across CIFAR-10/100 and TinyImageNet in highly non-IID settings. - Developed SurfScaff3D, a training-free text- and image-to-3D framework built on hierarchical B-spline surface patches for editable, open-domain 3D object generation.	
IIT Hyderabad Machine Learning & Vision Group, Dept. of CSE <i>Research Intern (Advisor: Prof. Vineeth N Balasubramanian, with Microsoft Research)</i>	Hyderabad, India May 2025 – Dec 2025
- Investigated concept quality in CBMs by analyzing concept locality to diagnose concept leakage; processing 2TB+ medical imaging data and applying locality-to-concept specific metrics. - Found that CBM models attend 45% more to spurious background regions than diagnostically relevant features, exposing a key medical-imaging failure mode.	
IIT Hyderabad Data-Driven Intelligence & Learning Lab, Dept. of AI <i>Research Intern (Advisor: Dr. Konda Reddy Mopuri)</i>	Hyderabad, India May 2024 – Aug 2024
- Benchmarked post-hoc explainability techniques (attention rollout, token pruning) across Vision Transformer architectures to quantify interpretability-efficiency tradeoffs. - Implemented token pruning strategies using top-K token overlay and patch importance metrics, reducing inference compute by 66% while keeping classification accuracy within 1-3% of the baseline.	

PUBLICATIONS

- Anay Loya*, Rishi Gupta*, **Arnav Samal**, and Anirban Chakraborty. “SurfScaff3D: Bridging Geometric and Semantic Representations for LLM-based 3D Object Generation.” *WACV, 2027 (Under Review)*

PROJECTS

Explainable Sketch to Face Generation – NIT Rourkela, Research Project Code	Aug 2025 – Nov 2025
- Fine-tuned Stable Diffusion v1.5 using ControlNet and LoRA, achieving an FSIM of 0.7386 to outperform baseline architectures in structural fidelity. - Engineered a temporal XAI framework adapting Grad-CAM to regression U-Nets, mapping gradient attribution across DDIM sampling timesteps to visualize feature-anchored sketch-to-face reasoning.	
Capsule Vision Challenge 2024 – Global Competition Preprint, Code	Aug 2024 – Oct 2024
- Architected a multi-model CNN and Vision Transformer ensemble pipeline, mitigating severe class imbalance via focal loss optimization and weighted sampling strategies. - Secured global Rank 5 out of 500+ participating teams.	